AU Property Insights

Power BI Dashboard — Data Connection Documentation

Task 8 — BI Advanced

File: AUProperty_PowerBI_Dashboard.pbix

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1. Overview

The AU Property Insights dashboard connects directly to the AUPropertyDW data warehouse — the star-schema database built in Task 6 on top of the staging/cleansing work from Task 5. All ten report tables (five dimensions, five facts) are sourced from the warehouse's star schema, loaded into Power BI using Import (in-memory) storage mode.

Property	Value
Source system	Microsoft SQL Server
Server / instance	SiriLenovo\IC
Database	AUPropertyDW
Schema used	star
Connector	Get Data → SQL Server database
Authentication	Windows Authentication (integrated), current user credentials
Storage mode	Import
Total tables loaded	10 (5 dimension, 5 fact)
Total relationships	9
Total measures	11

2. How the Connection Was Set Up

In Power BI Desktop:

- Home ribbon → Get Data → SQL Server database.
- Server: SiriLenovo\IC. Database: AUPropertyDW.
- Data Connectivity mode: Import (selected instead of DirectQuery so all visuals run against an in-memory cache for fast interactive performance).

- In the Navigator window, every table under the star schema was selected: DimSuburb, DimSchool, DimTransportStop, DimOffenceType, DimRentalHouseType, FactHouseValue, FactRentalValue, FactSchoolPresence, FactTransportAccess, FactCrime.
- Load was chosen directly (no Power Query edits were required beyond one text-cleanup step, described in Section 4).

3. Tables Imported

Table	Type	Columns	Purpose
DimSuburb	Dimension	8	Central hub dimension — suburb, city, state, postcode, latitude/longitude. All five fact tables relate to this table.
DimSchool	Dimension	8	School address, name, type and postcode/state details for FactSchoolPresence.
DimTransportStop	Dimension	7	Transport stop name, mode and location details for FactTransportAccess.
DimOffenceType	Dimension	4	Offence category/subcategory lookup for FactCrime.
DimRentalHouseType	Dimension	3	Rental property type lookup for FactRentalValue.
FactHouseValue	Fact	8	One row per house listing: house value, suburb key, load date.
FactRentalValue	Fact	8	One row per rental listing: weekly rental amount, rental house type key, suburb key.
FactSchoolPresence	Fact	5	Factless fact — one row per school-in-suburb presence record.
FactTransportAccess	Fact	5	Factless fact — one row per transport-stop-in-suburb access record.
FactCrime	Fact	7	One row per recorded incident: offence type key, suburb key, incident count.

4. Query Transformations Applied

Because the cleansing and conforming work was already completed in the Task 5/6 ETL pipeline, only one Power Query (M) transformation step was needed inside Power BI itself:

- DimSuburb — a Replaced Value step corrects a text-encoding artefact in the City column, where "Alice Springs" was stored as "?Alice Springs" for a small number of rows.

No other tables required M-level cleansing; all remaining shaping (bucket columns, sort-order columns, measures) was implemented natively in the Power BI data model as calculated columns and DAX measures rather than in Power Query, so that the underlying warehouse tables stay untouched.

5. Sample Power Query (M) Code

Every table follows the same Sql.Database(...) → Source{[Schema="star",Item="<TableName>"]}[Data] pattern. Two representative examples:

5.1 DimSuburb (includes the text-cleanup step)

```
let
   Source = Sql.Database("SiriLenovo\IC", "AUPropertyDW"),
   star_DimSuburb = Source{[Schema="star",Item="DimSuburb"]}[Data],
   #"Replaced Value" = Table.ReplaceValue(star_DimSuburb,
      "?Alice Springs", "Alice Springs", Replacer.ReplaceText, {"City"})
in
   #"Replaced Value"
```

5.2 FactHouseValue (straight load, no transformation)

```
let
   Source = Sql.Database("SiriLenovo\IC", "AUPropertyDW"),
   star_FactHouseValue = Source{[Schema="star",Item="FactHouseValue"]}[Data]
in
   star_FactHouseValue
```

6. Data Model Relationships

All nine relationships converge on DimSuburb, which acts as the central hub across all five fact tables — the fact-constellation design carried over from the Task 6 star schema. Every relationship is Many-to-One (fact → dimension), single-direction cross-filtering, and active.

From Table	From Column	To Table (dimension)	To Column
FactHouseValue	SuburbKey	DimSuburb	SuburbKey
FactRentalValue	SuburbKey	DimSuburb	SuburbKey
FactRentalValue	RentalHouseTypeKey	DimRentalHouseType	RentalHouseTypeKey
FactSchoolPresence	SuburbKey	DimSuburb	SuburbKey
FactSchoolPresence	SchoolKey	DimSchool	SchoolKey
FactTransportAccess	SuburbKey	DimSuburb	SuburbKey
FactTransportAccess	StopKey	DimTransportStop	StopKey
FactCrime	SuburbKey	DimSuburb	SuburbKey
FactCrime	OffenceTypeKey	DimOffenceType	OffenceTypeKey

7. Data Refresh Configuration

Because the model uses Import storage mode, report data is a snapshot loaded at the time of refresh rather than a live query against AUPropertyDW.

- Manual refresh: Home ribbon → Refresh in Power BI Desktop re-runs every table's M query against the live AUPropertyDW warehouse and reloads the in-memory model.
- Scheduled refresh: not configured for this submission, since the file has not been published to the Power BI Service. If published, scheduled refresh would require an on-premises data gateway, since SiriLenovo\IC is a local SQL Server instance rather than a cloud database.

Note: Auto date/time was disabled and any auto-generated date hierarchy tables were removed from the model, so the 10 tables listed in Section 3 are the complete and only tables in the model.

8. Calculated Columns and Measures Created

No relationships needed correction — Power BI auto-detected all nine SuburbKey/StopKey/SchoolKey/OffenceTypeKey/RentalHouseTypeKey relationships correctly on load, each Many-to-One, single-direction, and active (see Section 6). All analytical logic below was added directly in the Power BI data model.

8.1 Calculated Columns (4)

Two pairs of calculated columns were added — one text "bucket" column per value type for use as an axis/legend, plus a matching hidden numeric "sort" column (set as that column's Sort by Column) so the bucket labels display in logical price order instead of alphabetical or count order.

```
// FactHouseValue[HouseValue Bucket]
SWITCH(
    TRUE(),
    FactHouseValue[HouseValue] < 300000, "Under $300K",
    FactHouseValue[HouseValue] < 450000, "$300K - $450K",
    FactHouseValue[HouseValue] < 600000, "$450K - $600K",
    FactHouseValue[HouseValue] < 800000, "$600K - $800K",
    FactHouseValue[HouseValue] < 1200000, "$800K - $1.2M",
    FactHouseValue[HouseValue] < 2000000, "$1.2M - $2M",
    "Above $2M"
)
```

FactHouseValue[HouseValue Bucket Sort] mirrors this with the same thresholds returning integers 1–7 (hidden from the field list, used only as the Sort by Column). FactRentalValue[RentalValue Bucket] and [RentalValue Bucket Sort] follow the identical pattern on FactRentalValue[RentalAmount], with weekly-rent thresholds ($250 / $350 / $450 / $600 / $900 / $1,500).

8.2 Measures (11)

Measure	Table	DAX
Avg House Value	FactHouseValue	AVERAGE(FactHouseValue[HouseValue])
Median House Value	FactHouseValue	MEDIAN(FactHouseValue[HouseValue])
House Value Listings	FactHouseValue	COUNTROWS(FactHouseValue)
Avg Rental Amount	FactRentalValue	AVERAGE(FactRentalValue[RentalAmount])
Rental Listings	FactRentalValue	COUNTROWS(FactRentalValue)
School Count	FactSchoolPresence	COUNTROWS(FactSchoolPresence)
Distinct Schools	FactSchoolPresence	DISTINCTCOUNT(FactSchoolPresence[SchoolKey])
Transport Stop Count	FactTransportAccess	COUNTROWS(FactTransportAccess)
Distinct Stops	FactTransportAccess	DISTINCTCOUNT(FactTransportAccess[StopKey])
Total Recorded Incidents	FactCrime	SUM(FactCrime[RecordedIncidents])
Crime Categories	FactCrime	DISTINCTCOUNT(FactCrime[OffenceTypeKey])

Each COUNTROWS/DISTINCTCOUNT measure counts records or keys rather than summing a value, since FactSchoolPresence and FactTransportAccess are factless fact tables (presence/access records with no numeric measure of their own), and "listing counts" for house/rental value are naturally a row count rather than a sum.